

Name: Nguyễn Khánh Ly

Student_ID: 11247193

Class: DS66B

WINDOW FUNCTION

Lab 1: Film Ranking (The Basics)

WITH RankedFilms AS(

```
SELECT c.name AS category, f.title AS film_title, f.length AS film_len,  
RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC, f.title) AS rn  
FROM film f  
JOIN film_category fc ON f.film_id = fc.film_id  
JOIN category c ON c.category_id = fc.category_id
```

)

SELECT * FROM RankedFilms WHERE rn <= 3;

	category character varying (25) 🔒	film_title character varying (255) 🔒	film_len smallint 🔒	rn bigint 🔒
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Saturn Memo	182	2

Lab 2: Growth Analysis (Time Series)

WITH monthly_revenue AS (

SELECT

```
DATE_TRUNC('month', payment_date) AS revenue_month,  
SUM(amount) AS current_revenue
```

```

FROM
    payment
GROUP BY DATE_TRUNC('month', payment_date)
),

revenue_lag AS (
    SELECT
        revenue_month,
        current_revenue,
        LAG(current_revenue) OVER (ORDER BY revenue_month) AS previous_revenue
    FROM
        monthly_revenue
)

```

```

SELECT
    revenue_month,
    current_revenue,
    previous_revenue,
    ROUND(
        ((current_revenue - previous_revenue) / previous_revenue) * 100,
        2
    ) AS mom_growth_pct
FROM
    revenue_lag
ORDER BY
    revenue_month;

```

	revenue_month timestamp without time zone	current_revenue numeric	previous_revenue numeric	mom_growth_pct numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value (Running Total)

```

SELECT
    payment_id,
    customer_id,
    payment_date,
    amount,
    SUM(amount) OVER (PARTITION BY customer_id ORDER BY payment_date) AS running_clv
FROM payment
ORDER BY customer_id, payment_date;

```

	payment_id [PK] integer	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	running_clv numeric
1	18495	1	2007-02-14 23:22:38.996577	5.99	5.99
2	18496	1	2007-02-15 16:31:19.996577	0.99	6.98
3	18497	1	2007-02-15 19:37:12.996577	9.99	16.97
4	18498	1	2007-02-16 13:47:23.996577	4.99	21.96
5	18499	1	2007-02-18 07:10:14.996577	4.99	26.95
6	18500	1	2007-02-18 12:02:25.996577	0.99	27.94
7	18501	1	2007-02-21 04:53:11.996577	3.99	31.93
8	22680	1	2007-03-01 07:19:30.996577	4.99	36.92
9	22681	1	2007-03-02 14:05:18.996577	3.99	40.91
10	22682	1	2007-03-02 16:30:04.996577	0.99	41.90
11	22683	1	2007-03-17 11:06:20.996577	4.99	46.89
12	22684	1	2007-03-18 02:25:55.996577	0.99	47.88
13	22685	1	2007-03-19 08:23:42.996577	0.99	48.87

Lab 4: The "Top-N" Filter Challenge

WITH customer_spending AS (

SELECT co.country, c.customer_id, c.first_name, c.last_name, SUM(p.amount) AS total_spend,

RANK() OVER (PARTITION BY co.country ORDER BY SUM(p.amount) DESC) AS rnk

FROM customer c

JOIN payment p ON c.customer_id = p.customer_id

JOIN address a ON c.address_id = a.address_id

JOIN city ci ON a.city_id = ci.city_id

JOIN country co ON ci.country_id = co.country_id

GROUP BY co.country, c.customer_id, c.first_name, c.last_name

)

SELECT

country,

customer_id,

first_name,

last_name,

total_spend

FROM customer_spending

WHERE rnk = 1

ORDER BY country;

	country character varying (50) 🔒	customer_id integer 🔒	first_name character varying (45) 🔒	last_name character varying (45) 🔒	total_spend numeric 🔒
1	Afghanistan	218	Vera	Mccoy	67.82
2	Algeria	176	June	Carroll	151.68
3	American Samoa	320	Anthony	Schwab	47.85
4	Angola	528	Claude	Herzog	93.80
5	Anguilla	381	Bobby	Boudreau	99.68
6	Argentina	560	Jordan	Archuleta	129.71
7	Armenia	41	Stephanie	Mitchell	118.75
8	Austria	266	Nora	Herrera	113.74
9	Azerbaijan	334	Raymond	Mcwhorter	108.75
10	Bahrain	590	Seth	Hannon	108.76
11	Bangladesh	21	Michelle	Clark	146.68
12	Barbados	111	Olivia	St	100.10